

Regulatory Responses to Cryptoderivatives in the UK and the EU: The Future of Cryptoderivatives in the US

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Abstract

While regulators in the UK and the EU have recently moved to protect retail investors from the risk and volatility of the cryptoderivatives market, the US has taken a more hands-off approach. The US Congress should abandon this *laissez-faire* attitude toward cryptoderivatives and enact a comprehensive cryptocurrency regulation (“cryptoregulation”) that recognizes the novelty of cryptocurrencies’ market risks and introduces an effective regulatory infrastructure to curb market manipulation in the cryptocurrency spot market vis-à-vis the cryptoderivatives market. Comprehensive cryptoregulation would include (1) centralization of cryptocurrency trading platforms, (2) a mandatory registration requirement for all cryptocurrency exchanges, and (3) a federal cryptocurrency agency. This agency would have exclusive jurisdiction over cryptocurrencies in the US, including the spot market and any financial instruments where the underlying asset is a cryptocurrency (such as cryptoderivatives). A uniform federal-level cryptocurrency agency is likely to establish effective oversight and supervisory authority over the cryptocurrency spot markets, which would help curb market manipulation and restore investor confidence.

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What I'm concerned about at the moment is if it can be reasonably demonstrated that the underlying trading is generally not manipulated, it's happening on reliable venues with good rules...

- Jay Clayton, Chairman, Securities and Exchange Commission¹

I. Introduction

On October 6, 2020, the UK Financial Conduct Authority (“FCA”) prohibited the sale of cryptocurrency-related derivatives (“cryptoderivatives”) to retail investors on the ground that cryptocurrency as a reference asset is an unreliable basis for valuation for these derivatives products.² The FCA concluded that cryptoderivatives, especially in the form of contract for difference (“CFD”) and exchange-traded notes (“ETNs”), are ill-suited for retail consumers due to the harm they pose.³ In Europe, the European Securities and Markets Authority (“ESMA”) has also been looking to curb cryptoderivatives trading as these products are risky,

¹ SEC Chairman in response to investor protection regarding bitcoin ETF in Helen Partz, ‘SEC Chairman Highlights Investor Protection in Regard to Bitcoin ETF’ *The Cointelegraph* (14 March 2019) <<https://cointelegraph.com/news/sec-chairman-highlights-investor-protection-in-regard-to-bitcoin-etf>> accessed 11 November 2020.

² The FCA prohibiting the sale to retail clients of investment products that reference cryptoassets (FCA Final Rule) [2020] PS20/10.

³ Ibid.

speculative, and expose consumers to potentially huge losses.⁴ Both regulators seem to have three main reasons for banning the sale of cryptoderivatives to retail investors: first, cryptocurrencies' extremely volatility as a reference asset; second, the prevalence of rampant market abuse, price manipulation, and security breaches in the cryptocurrency spot market; and third, investors' significant lack of understanding of these complex derivatives products.⁵

While regulators have initiated a broader crackdown in the UK and the EU to protect retail investors from the cryptoderivatives markets' abuse and manipulation, US regulators chose to go in the opposite direction. In 2014, the Commodity Futures Trading Commission ("CFTC") approved TeraExchange, a bitcoin derivatives exchange, to self-certify bitcoin swaps allowing investors to trade dollar-dominated bitcoin currency swaps.⁶ The following year, the CFTC classified bitcoin as a commodity in its order against Coinflip, Inc.—a bitcoin trading platform, and thus ensured its entrance into the traditional derivatives market just like other commodities.⁷ Since then, several cryptoderivatives have proliferated in the market—the Chicago Mercantile Exchange ("CME") and Chicago Board Options Exchange ("CBOE") first launched cash-settled bitcoin futures in December 2017.⁸ The Intercontinental Exchange ("ICE") introduced physically-settled bitcoin futures and bitcoin options in September and

⁴ Silla Brush, 'U.K. FCA Bans Sale of Crypto-Derivatives to Retail Investors' *The Bloomberg* (New York, 6 October 2020) <<https://www.bloomberg.com/news/articles/2018-01-18/cryptocurrency-derivatives-may-fall-under-eu-retail-sales-curbs>> accessed 11 November 2020.

⁵ Previously, South Korea banned bitcoin futures trading following its initial ban on Initial Coin Offerings (ICOs) in 2017. See David Dinkins, 'In Unexpected Move, South Korean Regulator Suddenly Bans Bitcoin Futures Trading' 7 *The Cointelegraph* December 2017 <<https://cointelegraph.com/news/inunexpected-move-south-korean-regulator-suddenly-bans-bitcoin-futures-trading>> accessed 23 November 2020.

⁶ Michale Casey, 'TeraExchange Unveils First U.S. Regulated Bitcoin Swaps Exchange' *The Wall Street Journal* (New York, 12 September 2014) <<https://www.wsj.com/articles/teraexchange-launches-bitcoin-derivatives-exchange-1410543989>> accessed 11 November 2020.

⁷ *CFTC vs. Coinflip Inc.* [2015] CFTC Docket No. 15-29. (The CFTC held that bitcoin and other virtual currencies fall within §1a(9) of the Commodity Exchange Act as the definition of "commodity" shall include "all services, rights, and interests in which contracts for future delivery are presently or in the future dealt in." And therefore, any company offering bitcoin derivatives must comply with the CFTC laws, rules, and regulations.).

⁸ Evelyn Cheng, 'Bitcoin Debuts on the World's Largest Futures Exchanges, and Prices Fall Slightly' *The CNBC* (18 December 2019) <<https://www.cnbc.com/2017/12/17/worlds-largest-futures-exchange-set-to-launch-bitcoin-futures-sunday-night.html>> accessed 11 November 2020.

October 2019, respectively.⁹ With the CFTC's announcement that the cryptocurrency 'Ethereum' is a commodity,¹⁰ Eris Exchange ("ErisX") launched Ethereum-based physically settled futures contracts on May 11, 2020.¹¹

As of today, the US cryptoderivatives and perpetual swap market cap stands at \$48.58 billion.¹² As the market grows, the cryptoderivatives market's concerns are also emerging as unregulated online exchanges and brokerage firms offering cryptocurrency trading products are susceptible to spot market¹³ manipulation¹⁴ and cyber-attacks.¹⁵ To support the cryptoderivatives market,

⁹ Ryan Brown, 'NYSE Owner ICE Launches Deliverable Bitcoin Futures Contracts' *The CNBC* (23 September 2019) <<https://www.cnbc.com/2019/09/23/nyse-owner-ice-launches-deliverable-bitcoin-futures-contracts.html>> accessed 11 November 2020.

¹⁰ William Foxley, 'CFTC Chairman Confirms Ether Cryptocurrency is a Commodity' *The Coindesk* (10 October 2019) <<https://www.coindesk.com/cftc-chairman-confirms-ether-cryptocurrency-is-a-commodity>> accessed 11 November 2020.

¹¹ Nikhilesh De, 'ErisX Announces Launch of First US Ether Futures Contracts' *The Coindesk* (11 May 2020), <<https://www.coindesk.com/erisx-announces-launch-of-first-us-ether-futures-contracts>> accessed 11 November 2020.

¹² CoinMarket Cap 'Cryptocurrency Derivatives and Perpetual Swap Markets' <<https://coinmarketcap.com/derivatives/>> accessed 11 November 2020.

¹³ Spot market refers to the trading and prices of cryptocurrency in any exchange. Unlike traditional spot markets for commodities, cryptocurrencies have their own niche spot markets. There are over 300 cryptocurrency exchanges, and some Over the Counter markets that constitute the cryptocurrency spot markets.

¹⁴ Few notable examples of market manipulation trends in the cryptocurrency spot market: On October 20, 2020, the CFTC the Pennsylvania Federal Court orders a person to pay \$7.4 million for committing a multi-million-dollar bitcoin fraud (CFTC Release No. 8272-20). On June 18, 2019, the CFTC charged Control-Finance Limited—a purported bitcoin trading and investment company, and its principal, Benjamin Reynolds, for fraudulently obtaining and misappropriating \$147 million worth bitcoins from more than 1,000 customers (CFTC Release No. 7938-19). On July 23, 2018, the Federal Court orders a commodity pool operator and its principal to pay more than \$1.9 million in connection with a bitcoin and binary options fraud scheme (CFTC Release No. 7760-18). In another case, a New York Federal Court ordered a trading firm and its CEO to pay more than \$2.5 million for operating a bitcoin ponzi scheme (CFTC Release No. 7831-18).

¹⁵ In a study about market manipulation behavior in the cryptocurrency exchanges, the evidence shows that the biggest cryptocurrency exchange, Mt. Gox, was engaged in bitcoin price manipulation before it was hacked in 2013 that led the exchange to file for bankruptcy in 2014. A subsequent data leak revealed that a significant number of trades took place at rates that were far higher or far lower than the reference price. The findings also demonstrate that these abnormal transactions took place between two accounts (presumably belonged to Mt. Gox itself) which artificially inflated the daily bitcoin trade volume to manipulate the price. See Weili Chen, Jun Wu, Zibin Zheng, Chuan Chen, and Yuren Zhou 'Market Manipulation of Bitcoin: Evidence from Mining the Mt. Gox Transaction Network' (IEEE International Conference on Computer Communications, January 2019), <<https://arxiv.org/abs/1902.01941>> (accessed 11 November 2020); see also, Neil Gandal, JT Hamrick, Tyler Moore, and Tali Oberman, 'Price Manipulation in the Bitcoin Ecosystem' [2018] 95 *Journal of Monetary Economics* 86.

many argue that cryptoderivatives give institutional investors an efficient and confident way to hedge risk. However, this claim could be far from reality as two major global regulators—the FCA and the ESMA view cryptoderivatives as harmful to retail investors due to its opaque and uncertain nature. From a regulatory standpoint, the CFTC’s approach to regulating cryptoderivatives through self-certification (like traditional derivatives products) inadequately encapsulates the cryptocurrency spot market’s inherent risks of opacity, price volatility, and its exposure to market manipulation.¹⁶ Such inadequacy is embedded in the CFTC’s two opposite positions. In a traditional commodity derivatives market, the CFTC has the power and capacity to oversee the commodity spot markets and, therefore, to take enforcement actions against any abusive and manipulative behavior that is detrimental to the investors’ interests.¹⁷ However, concerning cryptoderivatives, the CFTC’s oversight mechanism over the cryptocurrency spot market is debatable as the CFTC, on repeated occasions, has appeared to have conflicting opinions regarding such power. Further, market participants in the spot market operate, assuming that the spot market is beyond the CFTC’s regulatory perimeter.¹⁸ Hence, in the absence of any regulatory clarity and the CFTC’s questionable oversight mechanism, the spot market could be a means to incentivize a bad actor to jeopardize cryptoderivatives markets’ integrity and thereby undermine retail investors’ confidence.

In addition, investor protection is a grey area in the US cryptocurrency regulatory regime. As an example, although an Initial Coin Offering (“ICO”) is a security,¹⁹ it is uncertain that all investments are protected under the Security Investment Protection Act (“SIPA”) given that the SEC has also determined that not all digital tokens are securities as depending on the degree of decentralization, a coin or token may fall outside the definition of a security.²⁰ Similarly,

¹⁶ Lee Reiners, ‘Bitcoin Futures: Self-certification to System Risk’ [2019] 23 N.C. Bank. Ins. 61.

¹⁷ Post financial crisis of 2007, the Dodd-Frank Act authorizes the CFTC to bring the OTC under a broader regulatory purview, and thereby establish a direct visibility into the commodity spot market.

¹⁸ See below, note 92.

¹⁹ The Security and Exchange Commission (SEC), ‘Report of Investigation Pursuant to Section 21(a) of the Securities Exchange Act of 1934: The DAO’ Release No. 81207 (25 July 2017) (“The DAO Report”).

²⁰ The DAO Report. The SEC is of the view that “[w]hether or not a particular transaction involves the offer and sale of a security—regardless of the terminology used—will depend on the facts and circumstance, including the economic realities of the transaction.” (pp.17-18). See also William Hinman, ‘Digital Asset Transactions: when

investor protection in the cryptoderivatives market also remains vague as the answer to the question lies in the effectiveness of the CFTC's regulatory and oversight mechanisms in preventing manipulation in the cryptocurrency spot market. Most investors, especially retail investors, lack an understanding of the complexity of cryptocurrency pricing and thus tend to treat cryptocurrency trading like gambling.²¹ Further, the complex pricing combined with extreme price volatility gives main-street investors an incentive to speculate the cryptocurrency price.²² Finally, despite cryptocurrency's market size, it incentivizes the institutional money to flow into the new cryptocurrency-based economy, and therefore, calls for regulators' vigilance.²³

Against this background, this paper puts forth a comparative analysis of the US regulatory responses to cryptoderivatives with specific references to the UK's and the EU's approaches and motives towards cryptoderivatives regulations in their respective regions. It discusses that the UK and the EU regulators primarily focus on protecting the retail investors from monetary losses arising from investment in cryptoderivatives products. In contrast, the US regulatory efforts are limited to interpreting cryptocurrency in light of the existing legal and regulatory framework, despite cryptocurrency's novel risks of price volatility and susceptibility to spot market manipulation, and the CFTC's lack of oversight over the cryptocurrency spot market. It explores the possibility of imposing an outright ban on cryptoderivatives like the UK but concludes that an outright ban on the cryptoderivatives market is likely to jeopardize financial

Howey Met Gary (Plastic)' 14 June 2018 <<https://www.sec.gov/news/speech/speech-hinman-061418>> accessed 23 November 2020.

²¹ In a market survey conducted in UK, the majority's perception regarding cryptocurrency is it is akin to betting. A study shows that cryptocurrency trading is linked with problematic gambling. See Yessi Bello Perez, 'Problem Gamblers More Likely to Obsessively Trade Cryptocurrency, Research Finds' *TheNextWeb.com* [BLOG], (Amsterdam, 11 March 2019) <Retrieved from <http://eproxy.lib.hku.hk/login?url=https://www-proquest-com.eproxy.lib.hku.hk/docview/2253326461?accountid=14548>> accessed 11 November 2020.

²² The cryptocurrency hedge-funds are betting on bitcoin's price. See Vincent Mislos, 'Bitcoin Price will Hit \$20,000 This Year because of 'Liquidity Pump', Says Novogratz' 30 July 2020 <<https://www.ibtimes.com/bitcoin-price-will-hit-20000-year-because-liquidity-pump-says-novogratz-3019421>> accessed 23 November 2020.

²³ Jason Grant Allen and Rosa María Lastra, 'Border Problems: Mapping the Third Border' (2020) *Modern Law Review* <doi:10.1111/1468-2230.12506> accessed 27 November 2020.

innovation growth in the US.²⁴ Furthermore, some cryptocurrency trading platforms are already complying with the existing laws and regulations, and an outright ban will put set them back. Therefore, to protect market integrity and safeguard retail investors' interest, the paper proposes that Congress enact a comprehensive cryptocurrency regulation ("cryptoregulation") recognizing the novelty of the cryptocurrencies' market risks and introducing effective regulatory treatments to curb market manipulation in the cryptocurrency spot market vis-à-vis cryptoderivatives market. This paper envisions a comprehensive cryptoregulation, which would include (1) centralization of cryptocurrency trading platforms, (2) a mandatory registration requirement for all cryptocurrency exchanges, and (3) a federal cryptocurrency agency, having exclusive jurisdiction over cryptocurrencies and oversight authority on the cryptocurrency spot markets.

II. Regulatory Framework of Cryptoderivatives in the EU and the UK

Unlike the US, regulators worldwide are skeptical about cryptoderivatives—mainly because of their extreme volatility and speculative, high-leverage nature.²⁵ The complexity of cryptoderivatives' products and investors' lack of understanding regarding the risks associated with them come with a high likelihood of losing money.²⁶ Therefore, among other major regulators, the EU and the UK have evaluated their regulations of cryptoderivatives on the ground that investment in these products hurt retail investors.²⁷

²⁴ In October 2019, the CFTC Chairman Health P. Tarbert expressed his intention to make the US a leading nation in the field of blockchain and digital assets. So, it is highly unlikely that the US regulators will take any decision of putting an outright ban on the cryptoderivatives.

²⁵ For instance, in May 2019, Japan asked bitFlyer to reduce leverage for its perpetual swap product. Emmanuel Goh, 'Crypto Derivatives: A Corner of the Market or the Market Itself?' *The Coindesk* (16 November 2019), <<https://www.coindesk.com/crypto-derivatives-a-corner-of-the-market-or-the-market-itself>> accessed 11 November 2020.

²⁶ Id.

²⁷ Maria Nikolova, 'ESMA Clarifies Scope of CFD Restrictions in Relation to Rolling Spot Forex' *The Finance Feeds* (28 September 2018) <<https://financefeeds.com/esma-clarifies-scope-cfd-restrictions-relation-rolling-spot-forex/>> accessed 11 November 2020.

In the EU, the ESMA is the independent market supervisory and law enforcement authority for the EU member states.²⁸ The ESMA is responsible for promoting “supervisory convergence and the consistent application of market rules”²⁹ within the EU. The ESMA first stepped into the cryptocurrency world by expressing its view on cryptocurrency token sales, i.e., ICOs, in November 2017.³⁰ Though the ESMA’s statement regarding token sales was vague, the ESMA has quite a strong position in regulating the EU’s cryptoderivatives market.³¹ In a Call for Evidence Report³² issued in January 2018,³³ the ESMA announced that cryptoderivatives, which are in the form of CFDs³⁴ and BOs,³⁵ should be subject to strict legal scrutiny alleging that these derivatives products are speculative and volatile, exposing investors to potentially

²⁸ Philipp Maume and Mathias Fromberger, ‘Regulation of Initial Coin Offerings: Reconciling US and EU Securities Laws’ [2019] 19(2) *Ch. Intl. L.J.* 548 <<http://cjil.uchicago.edu/publication/regulation-initial-coin-offerings-reconciling-us-and-eu-securities-laws>> accessed 11 November 2020.

²⁹ *Id.*

³⁰ The European Securities and Markets Authority (The ESMA), ‘The ESMA alerts firms involved in Initial Coin Offerings (ICOs) to the need to meet relevant regulatory Requirements’ ESMA50-157-828 <<http://perma.cc/A4BP-9QS4>> accessed 11 November 2020.

³¹ Note 21 at 556.

³² Before issuing this report, the ESMA previously published a statement on December 15, 2017 stating that it was considering measures to “restrict the marketing, distribution or sale to retail clients of CFDs and prohibit the marketing, distribution or sale to retail clients of BOs.” The follow-up Call for Evidence Report’s is to “seek additional evidence from firms and investors on the expected impact of the measures under consideration”. See ESMA, ‘Call For Evidence: Potential Intervention Measures on Contracts for Differences and Binary Options to Retain Clients’ ESMA35-43-904 <https://www.esma.europa.eu/sites/default/files/library/esma35-43-904_call_for_evidence_-_potential_product_intervention_measures_on_cfds_and_bos_to_retail_clients.pdf> accessed 11 November 2020.

³³ ESMA, ‘ESMA Consults On Potential CFDs And Binary Options Measures To Protect Retail Investors’ (18 January 2018) <<https://www.esma.europa.eu/press-news/esma-news/esma-consults-potential-cfd-and-binary-options-measures-protect-retail>> accessed 11 November 2019.

³⁴ “A CFD is defined as “a derivative other than an option, future, swap, or forward rate agreement, the purpose of which is to give the holder a long or short exposure to fluctuations in the price, level or value of an underlying, irrespective of whether it is traded on a trading venue, and that must be settled in cash at the option of one of the parties other than by reason of default or other terminational event.” *Id.*

³⁵ A BO is defined as “a derivative that meets the following conditions: (a) it must be settled in cash or may be settled in cash at the option of one of the parties other than by reason of default or other terminational event; (b) it only provides for payment at its close-out or expiry; (c) its payment is limited to: (i) a predetermined fixed amount if the underlying of the derivative meets one or more predetermined conditions; and (ii) zero or another predetermined fixed amount if the underlying of the derivative does not meet one of more predetermined conditions.” *Id.*

significant monetary loss.³⁶ The ESMA further called for responses from market participants regarding possible measures to regulate cryptoderivatives.³⁷ After considering all the responses and concerns, the ESMA, according to Art. 40 of Markets in Financial Instruments Regulation (“MiFIR”)³⁸ adopted restrictive product intervention measures in relation to CFDs and BOs. The intervention measures include (1) a prohibition on the marketing, distribution, or sale of BOs and (2) a restriction on the marketing, distribution, or sale of CFDs to retail investors.³⁹ In adopting these restrictive measures, the ESMA noted:

“CFDs are complex products. The pricing, trading terms, and settlement of such products is not standardized, impairing retail investors’ ability to understand the terms of product...Retail investors find it difficult to understand and assess the expected performance of a CFD... Furthermore, the offer of CFDs to retail investors has increasingly featured aggressive marketing practices as well as misleading communications...”⁴⁰

It also noted that the cryptocurrency is an immature asset class that poses “separate and significant concerns.”⁴¹ Therefore, retail investors hardly understand the risk of speculation on cryptoderivatives products. The ESMA further extended its restriction on CFDs in September

³⁶ The ESMA, ‘Additional Information on the Agreed Product Intervention Measure Relating to Contract for Differences and Binary Options’ (27 March 2018) <https://www.esma.europa.eu/sites/default/files/library/esma35-43-1000_additional_information_on_the_agreed_product_intervention_measures_relating_to_contracts_for_differences_and_binary_options.pdf> accessed 11 November 2020.

³⁷ Id.

³⁸ Art. 40 of MiFIR permits the ESMA to temporarily prohibit, restrict marketing, distribution, or sale of certain financial instruments on grounds of investor protection and market integrity.

³⁹ The restrictions on BOs came in effect from 1 July 2018 whereas the restrictions on CFDs came in effect from 1 August 2018 (the restrictions on CFDs are renewable).

⁴⁰ Note 38 at p.2.

⁴¹ The ESMA also states: “CFD with cryptocurrencies as an underlying raise separate and significant concerns. Cryptocurrencies are a relatively immature asset class that pose major risks for investors [due to the specific characteristics of cryptocurrencies]... retail clients typical do not understand the risks involved when speculating on an extremely volatile and relative immature asset class, which are exacerbated by trading on margin, as it requires the client to react in a very short time period.” Id.

2019 for another three months.⁴² In the renewal notice, the ESMA stated that “significant investor protection concern related to the offer of CFDs to retail clients continues to exist.”⁴³

Similarly, the FCA, which regulates the UK financial services industry, has imposed strict regulatory measures in the UK cryptoderivatives market.⁴⁴ Before finalizing the outright ban on cryptoderivatives,⁴⁵ the FCA first proposed a temporary ban on cryptoderivatives and ETNs in 2019 on the ground that cryptoderivatives products were ill-suited to retail customers “who are unable to assess the value and risks of derivatives or ETNs reliably.”⁴⁶ To assess the trend of investors’ increasing interest and its correlation with cryptocurrencies’ price instability, the FCA evaluated the price of bitcoin and Ethereum, and Google trends data between 2018 and 2019, to demonstrate that retail investors’ interests are “correlated to the increasing price of bitcoin.”⁴⁷ The FCA is of the view that these data further demonstrate investors’ speculative behavior over a price-boom in cryptocurrency and do not necessarily investors’ ability to assess the intrinsic value of cryptocurrency reliably and consistently or the derivatives that use cryptocurrency as a reference asset.⁴⁸ Therefore, in framing the grounds for banning cryptoderivatives, the FCA’s central focus was protecting retail investors from monetary losses.⁴⁹ The regulators were concerned that retail investors could be hurt (i) by the opacity

⁴² The ESMA, ‘ESMA to Renew Restriction on CFD for a Further Three Months’ (28 September 2018) <https://www.esma.europa.eu/sites/default/files/library/esma71-99-1041_-_esma_to_renew_restriction_on_cfds_for_a_further_three_months.pdf> accessed 11 November 2020.

⁴³ “France : ESMA to Renew Restrictions on CFDs for a Further Three Months from 1 February 2019.” MENA Report, Albawaba (London) Ltd., Dec. 2018.

⁴⁴ In April 2018, the FCA released additional guidance regarding derivative contracts on cryptocurrencies, making it clear that derivatives on crypto tokens are transferable securities and that providing financial services in this regard require formal authorization. The FCA, ‘FCA proposes ban on sale of crypto-derivatives to retail consumers’ (3 July 2019) <<https://www.fca.org.uk/news/press-releases/fca-proposes-ban-sale-crypto-derivatives-retail-consumers>> accessed 11 November 2020.

⁴⁵ Note 2.

⁴⁶ The FCA applies Art. 42 of MiFIR to propose the ban on cryptoderivatives.

⁴⁷ FCA, ‘Prohibiting the Sale to Retail Clients of Investment Products that Reference Cryptoassets: Technical Annex’ (October 2020) <https://www.fca.org.uk/publication/policy/ps20-10.pdf> accessed 20 November 2020.

⁴⁸ Note 1 at p.8.

⁴⁹ The FCA, ‘Prohibiting the Sale to Retail Client of Investment Products That Reference Cryptoassets’ (July 2019), <<https://www.fca.org.uk/publication/consultation/cp19-22.pdf>> (accessed 11 November 2020).

and complexity of cryptocurrency as reference assets,⁵⁰ (ii) by retail consumers' lack of understanding and consequent inability to make an informed investment decision on cryptoderivatives,⁵¹ and (iii) because cryptocurrency as a reference asset is the highly speculative,⁵² volatile,⁵³ and susceptible to sudden price drops and abrupt price dislocation.⁵⁴ Therefore, allowing cryptoderivatives to grow in the retail market might create a perception among retail investors that these products are suitable and appropriate investment products.⁵⁵

During the interim phase, the FCA considered other “less interventionist” approaches—for instance, ‘do nothing’⁵⁶ or ‘provide further consumer warnings.’⁵⁷ Nevertheless, it concluded that “any remedy other than a ban on the sale to retail clients would fall short of adequately reducing the harms to consumers and risks identified.”⁵⁸ The FCA’s efforts to regulate cryptoderivatives was not unopposed.⁵⁹ The FCA’s position was challenged on the ground that

⁵⁰ Id at p.14.

⁵¹ Id.

⁵² Shay-Kee Tan, Jennifer So- Kuen Chan and Kok-Haur Ng, ‘On the Speculative Nature of Cryptocurrencies: A Study of Garman and Klass Volatility Measure’ [2020] 32 *Finance Research Letters* <<https://doi.org/10.1016/j.frl.2018.12.023>> accessed 16 November 2020.

⁵³ Bitcoin is 26 times as volatile than S&P500; see C. Baek and M. Elbeck ‘Bitcoin as an investment or speculative: A first look’ [2015] 22:1 *Applied Economics Letter* 34 <<https://doi.org/10.1080/13504851.2014.916379>> accessed 16 November 2020.

⁵⁴ Note 2.

⁵⁵ The FCA states: “In our view, these products do not meet a legitimate investment need and are not widely used for hedging purposes. Instead, retail customers use them for purely speculative measures. This risk may be increased by investors’ limited understanding of cryptoassets.” See Note 51 at p.21.

⁵⁶ The FCA is of the opinion that a do nothing” approach does not address the fundamental product flaws or address the significant harm to consumers posed by these products. Existing disclosure obligations and appropriateness tests are unlikely to be effective in conveying the risks to retail clients. Continuing to allow the offer of these products by firms with FCA authorization may also give retail investors a false sense of security by contrast to the underregulated nature of the underlying.

⁵⁷ The FCA, ‘Consumer warning about the risks of investing in cryptocurrency CFDs’ (03 July 2019) CP19/22 <<https://www.fca.org.uk/news/news-stories/consumer-warning-about-risks-investing-cryptocurrency-cfds>> accessed 11 November 2020.

⁵⁸ Note 51 at p.24.

⁵⁹ Steve Kaaru, ‘CoinShares Wants Users to Take Action Against UK Crypto Assets Ban’ *The Coingeek* (29 September 2019) <<https://coingeek.com/coinshares-wants-users-to-take-action-against-uk-crypto-assets-ban/>> accessed 11 November 2020.

“an outright ban would affect its members who are already in compliance with a slew of regulatory standards.”⁶⁰ However, the FCA continues to maintain its position on the matter to protect retail investors, stating, “a ban on cryptoderivatives could lead to a \$96 million haircut in harm done to retail traders per year.”⁶¹

III. Regulatory Framework of Cryptoderivatives in the US

In the US, the Commodity Exchange Act (“CEA”) and the Commodity Futures Trading Commission Rules (“CFTC Rules”) regulates the trading of derivatives contracts (futures, options, swaps), and the CFTC supervises the commodity and derivatives markets and. The CEA, which is the primary statute governing the laws and regulations of the US derivatives market, defines “commodity” to include agricultural products, “all other goods and articles,” and “all services, rights, and interests” in which “contracts for future delivery are presently or in the future dealt in.”⁶² In 2015, the CFTC assumed that certain virtual currencies, such as bitcoin and litecoin, are commodities and should be regulated by the CFTC.⁶³ Besides, multiple federal courts also held that virtual currencies are commodities as per CEA.⁶⁴ Hence, cryptoderivatives— such as bitcoin-futures, swaps, and options—fall within the CFTC’s regulatory perimeter.⁶⁵

⁶⁰ Osato Avan Nomayo, ‘Cryptoderivatives Ban: The UK Govt Won’t Interfere with FCA’ *The Blockonomi* (29 October 2019) <<https://blockonomi.com/crypto-derivatives-ban-uk-govt-wont-interfere-with-fca/>> accessed 11 November 2020.

⁶¹ *Id.*

⁶² The CEA, §1a (9).

⁶³ Note 14. For illustration, see also Matt Clinch, ‘Bitcoin Officially Becomes a Commodity’ *The CNBC* (15 September 2018) <<https://www.cnbc.com/2015/09/18/bitcoin-now-classed-as-a-commodity-in-the-us.html>> accessed 11 November 2020.

⁶⁴ *CFTC v. Coinflip, Inc.* [2015] CFTC Docket No. 15-29 <<http://www.cftc.gov/idc/groups/public/@lrenforcementactions/documents/legalpleading/enfcoinfliporder09172015.pdf>> accessed 11 November 2020. see also, *CFTC v. McDonnell* [2018] 287 F. Supp. 3d 213 (E.D.N.Y. 2018). See also, *CFTC v. My Big Coin Pay, Inc.* [2018] 334 F. Supp. 3d 492 (D. Mass. 2018).

⁶⁵ See Houman B. Shadab, ‘Regulating Bitcoin and Blockchain Derivative’ NYLS Legal Studies Research Paper (9 October 2020) <https://www.cftc.gov/sites/default/files/idc/groups/public/@aboutcftc/documents/file/gmac_100914_bitcoin.pdf> accessed 11 November 2020 (Discussing bitcoins fall within the definition of “commodity” under the Commodity Exchange Act (CEA) and therefore, derivatives contracts like futures, swaps, options that reference bitcoins are subject to regulation by the Commodity Futures Trading Commission (CFTC)).

In December 2017, the CFTC permitted futures exchanges to apply the self-certification process for bitcoin futures and binary options under §7(a)(2) of the CEA.⁶⁶ However, despite the CFTC's attempt to normalize cryptoderivatives in the existing legal and regulatory framework, these derivatives products pose numerous risks to retail consumers. Its lack of direct oversight on the cryptocurrency spot market poses a significant challenge to regulate market manipulation, which has adverse impacts on cryptoderivatives investors. Daniel Grofine, Director of the CFTC's fintech initiative ("LabCFTC"), shared similar concerns on the issue of cryptocurrencies and digital assets during his testimony before the US House Committee on Agriculture in July 2018.⁶⁷ He warned that while many things could be commodities, the CFTC's direct oversight on the commodity spot market is essential to bring those commodity-built futures, swaps, and options within its regulatory perimeter. The current regulatory approach should focus on bringing clarity and certainty to the market, and any "hasty regulatory pronouncements are likely to...have unintended consequences, or fail to capture important nuance regarding the structure of new products and models."⁶⁸

Currently, there are several cryptoderivatives products available to US retail customers.⁶⁹ The ICE launched its first bitcoin-settled futures—the Bakkt futures—in September 2019.⁷⁰ Three months later, the ICE introduced its monthly settled bitcoin options.⁷¹ In January 2020, the CME started trading options on bitcoin futures.⁷² ErisX launched ether-based physically settled futures contracts in May 2020.⁷³ In addition to bitcoin and ether derivatives products, the

⁶⁶ CFTC, 'A CFTC Primer on Virtual Currencies' (17 October 2017) <https://www.cftc.gov/sites/default/files/idc/groups/public/documents/file/labcfrc_primercryptocurrencies100417.pdf> accessed 12 November 2020.

⁶⁷ Id.

⁶⁸ Id.

⁶⁹ For reference, The CoinMarketCap <<https://cryptoderivatives.market/>> accessed 11 November 2020.

⁷⁰ Note 8.

⁷¹ Adam White, 'Expanding the Bakkt Bitcoin Product Complex: Bitcoin Options and Cash Settled Futures Now Available' *The Bakkt BLOG* (9 December 2019) <<https://medium.com/bakkt-blog/expanding-the-bakkt-bitcoin-product-complex-68000faea6b3>> accessed 11 November 2020.

⁷² The CME Group <<https://www.cmegroup.com/trading/bitcoin-futures.html>> accessed 11 November 2020.

⁷³ Note 11.

cryptocurrency industry will soon attempt to issue other cryptocurrency-based derivatives products. However, there are at least three regulatory issues with the CFTC's approach to approving these new cryptoderivatives.⁷⁴ First, the CFTC's traditional approach to regulating cryptoderivatives, primarily through the "self-certification" process, is risky as the existing legal framework of "self-certification" is not adequate to prevent price manipulation in the cryptocurrency spot market. Second, the CFTC's view on market manipulation in the cryptocurrency spot market contradicts the SEC's view on the same issue. And third, the CFTC is surprisingly numb to the suggestion that cryptoderivatives could jeopardize retail investors' interest, and such an approach deviates from the two major global regulators—namely, the ESMA and the FCA.

A. The CFTC's Self-Certification Process and Heightened Review for Cryptoderivatives Undermine Cryptocurrencies' Market Risks

The "self-certification" process for derivatives contracts was introduced in 2000 by enacting the Commodity Futures Modernization Act.⁷⁵ Under this new law, the CFTC permits the listing of a new futures contract if "(1) the exchange submits a written self-certification to the CFTC certifying that the contract complies with CEA and CFTC regulations, or (2) the exchange has voluntarily submitted the contract for CFTC approval."⁷⁶ Therefore, in the self-certification process, the exchanges themselves can verify that a new contract complies with CEA or CFTC requirements.⁷⁷ The designated contract markets ("DCMs") may also voluntarily submit new contracts for approval to the Commission and list the futures contract within twenty-four hours upon the CFTC's approval of the contract.⁷⁸ The CFTC's self-certification process has been

⁷⁴ This paper will not discuss systemic risk aspect of cryptoderivatives. But the paper admits that the cryptoderivatives connect regulated sectors, i.e., firms and financial institutions with the unregulated underlying cryptocurrency markets. Therefore, any contagion created in the unregulated asset class may have a spillover impact on the regulated sector, can give rise to systemic risk. For a detail discussion, see Note 16.

⁷⁵ The CEA §7(a)(2).

⁷⁶ Note 17.

⁷⁷ Id at 71.

⁷⁸ Id.

questionable since its introduction.⁷⁹ Between 2000 and 2017, data⁸⁰ suggest that the self-certification process facilitated the approval and listing of many complex exchange-traded commodity derivatives.⁸¹ Such approval process often included an absence of a proper understanding of the traded-products giving rise to opacity and unpredictability in the market, which potentially increased “inefficiency and system failure” across the financial system.⁸²

Despite criticisms of the self-certification process and its controversial role in the 2007 financial crisis, the CFTC allowed the process for cryptoderivatives in 2014 as TeraExchange self-certified its bitcoin non-deliverable forwards.⁸³ As of today, several US futures exchanges such as CME, CBOE, ICE, and ErisX self-certified both cash-settled and physically-settled cryptocurrency-based (bitcoin and ether) futures contracts and binary options.⁸⁴ Although the CFTC claimed that the Commission “held rigorous discussions” with the exchanges for weeks before allowing them to self-certify these cryptoderivatives products,⁸⁵ such a process spurred agitation among the futures industry.⁸⁶ Walt Lukken, CEO of the Futures Industry Association, expressed his concerns:

⁷⁹ See Saule T. Omarova, ‘Licence to Deal: Mandatory Approval of Complex Financial Products’ [2010] 90 Wash. U. L. Rev. 63.

⁸⁰ Note 16 at 72. The author compiled the data from a publicly available database on the CFTC’s website which clearly indicates a significant increase of the number of new exchange-traded products approved through self-certification process. It is also suggested that this might have potentially contributed to the financial crisis of 2007.

⁸¹ Although the economic purpose of the CFTC’s self-certification rule was to “reduce the potential threat of market manipulation or congestion,” during the financial crisis of 2008, the market could not necessarily extricate themselves from the underlying cash markets and the policy goal of preventing potential harm to such markets from excessive financial speculation.

⁸² For scholarly analyses, see Steven L. Schwartz, ‘Regulating Complexity in Financial Markets’ [2010] 87 Wash. U. L. Rev. 211.

⁸³ Note 6.

⁸⁴ David Felsenthal et al., ‘Clifford Chance Discusses the Role of the CFTC in the Regulation of Bitcoin’ *The CLS Blue Sky (BLOG)* (16 February 2018) <<http://clsbluesky.law.columbia.edu/2018/02/16/clifford-chance-discusses-the-role-of-the-cftc-in-the-regulation-of-bitcoin/>> accessed 13 November 2020.

⁸⁵ CFTC, ‘CFTC Statement on Self-Certification of Bitcoin Products by CME, CFE, and Cantor Exchange’ (1 December 2017) <<https://www.cftc.gov/PressRoom/PressReleases/pr7654-17>> accessed 13 November 2020.

⁸⁶ Walt Lukken, ‘Open letter to CFTC chairman Giancarlo regarding the listing of cryptocurrency derivatives’ *The Futures Industry Association* (7 December 2017) <<https://fia.org/articles/open-letter-cftc-chairman-giancarlo-regarding-listing-cryptocurrency-derivatives>> accessed 13 November 2020.

“We remain apprehensive with the lack of transparency and regulation of the underlying reference products on which these products are based and whether exchanges have the proper oversight to ensure the reference products are not susceptible to manipulation, fraud, and operational risk.”

Besides, the CFTC does not oversee the cryptocurrency spot market, making it more susceptible to fraud and price manipulation.⁸⁷ To respond to the FIA’s concern and provide more clarity, the CFTC came up with a stricter review for self-certified cryptoderivatives products, i.e., “heightened review,”⁸⁸ for all bitcoin futures and cryptoderivatives products that will apply through self-certification.⁸⁹ However, such a review is also questionable because the “heightened review” does not provide the CFTC with an effective oversight mechanism for the cryptocurrency spot market, and therefore cannot cure the risks of cryptoderivatives (analysis below).

B. ‘Heightened Review’ Does Not Provide the CFTC with the Desired Visibility into the Cryptocurrency ‘Spot’ Markets

Cryptocurrency spot markets operate in an unregulated space or with little regulatory clarity. For instance, ErisX, which offers Ethereum-based futures contract, claims that the ErisX spot market is beyond CFTC’s regulatory purview. It says:

The CFTC does not have regulatory oversight over virtual currency products including spot market trading of virtual currencies. ErisX spot market is not licensed, approved, or registered with the CFTC and transaction on the *ErisX Spot Market are not subject to CFTC rules, regulations or regulatory oversight*.⁹⁰ (Emphasis is mine).

⁸⁷ Id.

⁸⁸ “Heightened review is a new process, without statutory process, without statutory basis, that the CFTC is using to review new virtual currency derivative products.” Note 14 at 74.

⁸⁹ CFTC, ‘CFTC Backgrounder on Oversight of and Approach to Virtual Currency Futures (4 January 2018) <https://www.cftc.gov/sites/default/files/idc/groups/public/@newsroom/documents/file/backgrounder_virtualcurrency01.pdf> accessed 13 November 2020 (The CFTC Oversight Background Paper).

⁹⁰ ErisX <<https://www.erisx.com/about/investors/>> accessed 20 November 2020.

Under the CEA, the CFTC is mandated to prevent market manipulation in the derivatives market,⁹¹ which gives the CFTC an authority to act against the price manipulation of any underlying commodity. To achieve this goal, the CFTC intends to ensure that the self-certified cryptoderivatives contracts are not “readily susceptible to manipulation.”⁹² The heightened review also allows the CFTC to implement risk-mitigation and oversight mechanisms through heightened margin requirements and information-sharing agreements among cryptocurrency exchanges.⁹³ The CFTC views that the information-sharing agreements among cryptocurrency exchanges will ensure the CFTC’s access to data that could “facilitate the detection and pursuit of bad actors in the underlying spot market.”⁹⁴ However, unlike other traditional commodity spot markets, there is no US law providing “direct, comprehensive federal oversight of underlying bitcoin or virtual currency spot markets.”⁹⁵ Many of the platforms are located offshore and are not registered with the CFTC or the SEC. Therefore, the CFTC’s satisfaction that the “information-sharing agreements” would ensure their visibility into the cryptocurrency spot market is debatable.⁹⁶

Also, in cash-settled cryptocurrency futures, the ability to manipulate depends on “how easily the reference rate that is used to price the contract can also be manipulated.”⁹⁷ Though there is no evidence of price manipulation of the CME or ICE futures to date, such manipulation on

⁹¹ The CEA § 6(c)(1), § 6(c) (3) and § 9(1); CFTC, ‘Prohibition on the Employment, or Attempted Employment, of Manipulative and Deceptive Devices and Prohibition on Price Manipulation’ (14 July 2011) <<https://www.federalregister.gov/documents/2011/07/14/2011-17549/prohibition-on-the-employment-or-attempted-employment-of-manipulative-and-deceptive-devices-and>> accessed 13 November 2020 (Final Rule 180.1).

⁹² 17 C.F.R. section 38.200.

⁹³ The CFTC Oversight Background Paper.

⁹⁴ According to the CFTC Oversight Background Paper, heightened review includes, among others, DCMs to enter into “direct or indirect information sharing agreements with spot market platforms to allow access to trade and trader data.” See also Note 13.

⁹⁵ Jerry Brito, Houman B. Shadab, and Andrea Castillo, ‘Bitcoin Financial Regulations: Securities, Derivatives, Prediction Markets, and Gambling’ [2014] 51 Colum. Sci. & Tech L. Rev. 144 (Arguing that “physically-settled transactions are generally not subject to the full scope of CFTC regulation precisely because they do not implicate the markets that the CFTC is concerned about, namely, futures and swaps markets.”).

⁹⁶ Note 14 at 75.

⁹⁷ Id.

cryptocurrency exchanges is rampant.⁹⁸ In the absence of federal oversight, crypto-exchanges are widely involved in price tampering through creating fake trade volumes (commonly known as “pump-and-dump”).⁹⁹ In many instances, traders use social media to perform pump-and-dump schemes to inflate the virtual currencies’ prices artificially.¹⁰⁰ A recent study reveals Telegram and Discord’s large-scale pump-and-dump scheme.¹⁰¹ Such pumping-and-dumping activity can hurt investors in the cryptoderivatives markets in the long run because cryptoderivatives enable institutional investors (and also potential manipulators) to bet on the future bitcoin’s price. It is not unlikely that a group of traders would place a massive trade on a bitcoin spot market on the contract’s settlement date, thereby pushing up the price of bitcoin and earning a profit on the futures position; and in the same way, if the speculation is on a decreased price, instantly dump the trade.

Despite the apprehension, there is a regulatory vacuum in enforcing cryptocurrency pump-and-dump.¹⁰² Usually, the SEC enforces pump-and-dump schemes since it is a type of securities fraud.¹⁰³ Nevertheless, in the existing federal securities law, the SEC is not likely to intervene

⁹⁸ For reference, see note 14.

⁹⁹ Paul Vigna, ‘Most Bitcoin Trading Faked by Unregulated Exchanges: Study Finds: Bitwise’s research casts new doubts about fraud in nascent market’ *The Wall Street Journal* (New York, 22 March 2019) <https://www.wsj.com/articles/most-bitcoin-trading-faked-by-unregulated-exchanges-study-finds-11553259600?mod=hp_lead_pos7> accessed 13 November 2020. See Kate Rooney, ‘Majority of Bitcoin Trading is a Hoax’ *The CNBC* (23 March 2019) <<https://www.cnbc.com/2019/03/22/majority-of-bitcoin-trading-is-a-hoax-new-study-finds.html>> accessed 13 November 2020.

¹⁰⁰ JT Hamerick, Farhang Rouhi, Arghya Mukharjee, Amir Feder, Nei Gandai, Tyle Moore, and Marie Vasek, ‘An Examination of the Cryptocurrency pump and dump ecosystem’ (18 December 2018) <https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3303365> accessed 13 November 2020.

¹⁰¹ Michael Mckee, ‘Trader Using “Pump and Dump” Schemes to Manipulate Cryptocurrency Prices’, *The Finbrief (BLOG)* (22 August 2018) <<https://blogs.dlapiper.com/globalfinance/2018/08/22/traders-using-pump-and-dump-schemes-to-manipulate-cryptocurrency-prices/>> accessed 13 November 2020). (discussing “[s]uch pump and dump scheme are accomplished through private chatrooms which are accessible only by invitation, and generally overseen by an anonymous moderator. The strategy is to announce a date, time, and exchange for a pump of typically illiquid cryptocurrency. As the buying frenzy pushes the prices up, the members of the pump group begin dumping, i.e., selling at the signal. Successful traders gloat about their profits.”).

¹⁰² In a report published by the New York Office of the Attorney General admits that the regulators lack control to evade pump and dump activity. See New York State Office of the Attorney General, *Virtual Currency Integrity Initiative Report* (September 2018), <https://ag.ny.gov/sites/default/files/vmii_report.pdf> accessed 23 November 2020.

¹⁰³ The Exchange Act 1934, Rule 10-b(5).

as it has determined that bitcoin is not a security.¹⁰⁴ Hence, the question remains, will the CFTC oversee such illegal activities in the cryptoderivatives commodity spot market under its mandate of preventing fraud and market manipulation? In 2018, the CFTC first issued an advisory note to warn the consumers to beware and avoid pump-and-dump schemes occurred in cryptocurrency trading.¹⁰⁵ Nonetheless, the CFTC's overall approach does not adequately address the fraud and manipulation concerns in the cryptocurrency spot market.¹⁰⁶ On the other hand, in the absence of any oversight mechanisms, cryptocurrency spot markets operate in an unregulated space, which is likely to incentivize manipulative behavior in the cryptoderivatives market.

C. Cryptoderivatives Serve the Interest of Cryptocurrency Exchanges Only

Many scholars argue that cryptoderivatives would only serve the interest of the cryptocurrency exchanges because cryptoderivatives allow these exchanges to hedge their risk exposures that arise from the volatility in the cryptocurrency spot market.¹⁰⁷ If someone purchases a piece of furniture on Overstock and pays in bitcoin via Coinbase, the payment is denominated in dollars and transferred from Coinbase to Overstock's bank account. This means "it is Coinbase that is accepting the exchange volatility risk."¹⁰⁸ Even though Coinbase charges Overstock a certain percentage as a payment processing fee, such a fee is not sufficient to "cover the exchange rate risk that Coinbase could face in the future."¹⁰⁹ It will only make sense if Coinbase can hedge its exchange rate risk by "simply engaging in swap or futures contract."¹¹⁰ Arguably, this is

¹⁰⁴ Eugene Kim, 'The SEC warns investors about potential ICO scams and 'pump and dump' schemes' *The CNBC* (28 August 2017) <<https://www.cnbc.com/2017/08/28/sec-warns-on-ico-scams-pump-and-dump-schemes.html>> accessed 13 November 2020.

¹⁰⁵ CFTC, 'CFTC Warns Customers to Avoid Pump-and-Dump Schemes' (15 February 2018) <<https://www.cftc.gov/PressRoom/PressReleases/pr7697-18>> accessed 22 November 2020.

¹⁰⁶ In the past, the CFTC publicly announced that it is not the CFTC's duty to oversee a spot market on a daily basis. cited in note 14 at 85.

¹⁰⁷ Note 97.

¹⁰⁸ Cade Metz, 'The Grand Experiment Goes Live: Overstock.com is Now Accepting Bitcoin' *The Wired* (4 January 2014) <<https://www.wired.com/2014/01/overstock-bitcoin-live/>> (discussing the volatility risks associated with Bitcoin and Overstock's collaboration with Coinbase).

¹⁰⁹ Note 97 at p.157.

¹¹⁰ Id at 157-58.

one of the reasons why cryptocurrency exchanges were insisting on cryptoderivatives for so long.¹¹¹ Also, the cryptocurrency markets are already flooded with every kind of crime—fraud,¹¹² scams,¹¹³ hacks,¹¹⁴ insider trading¹¹⁵—and the hype is so overwhelming that retail investors often sell everything they have to buy digital currencies. Moreover, it is quite uncertain what impacts the market price of cryptocurrencies. Any regulatory move appears to impact the bitcoin price as we had seen that in the past, the bitcoin price dropped by 30% when China banned cryptocurrency or South Korea initiated a crackdown on cryptocurrency.¹¹⁶ In March 2020, when the US market was turbulent due to the COVID-19 crisis combined with a plummet in oil prices and sell-off in stocks,¹¹⁷ the cryptocurrency market lost almost \$26.43 billion in a day.¹¹⁸ In addition to spot markets, the abrupt price dips regularly wipes out billions of dollars from other cryptocurrency-based markets, such as the decentralized finance market.¹¹⁹

¹¹¹ Id.

¹¹² Collen Shalby, ‘Camarillo man and two others arrested in alleged \$722-million cryptocurrency fraud scheme’ *The Los Angeles Times* (Los Angeles, 10 December 2019) <<https://www.latimes.com/california/story/2019-12-10/camarillo-man-and-two-others-arrested-in-alleged-722-million-cryptocurrency-fraud-scheme>> accessed 13 November 2020.

¹¹³ Crypto Slate, ‘Twitter Bitcoin Scams Take New Leap After Verified Twitter Accounts Impersonate Elon Musk’ (18 November 2019) <https://cryptoslate.com/twitter-bitcoin-scams-take-new-leap-after-verified-twitter-accounts-impersonate-elon-musk/> accessed 13 November 2020.

¹¹⁴ Eric Lam, ‘Hackers Steal \$40 Million Worth of Bitcoin from Binance Exchange’ *The Bloomberg* (08 May 2019) <<https://www.bloomberg.com/news/articles/2019-05-08/crypto-exchange-giant-binance-reports-a-hack-of-7-000-bitcoin>> accessed 21 November 2020. Andrew Norry, ‘The History of the Mt. Gox Hack: Bitcoin’s Biggest Hit’ *The Blockonomi* (07 June 2019) <<https://blockonomi.com/mt-gox-hack/>> accessed 13 November 2020.

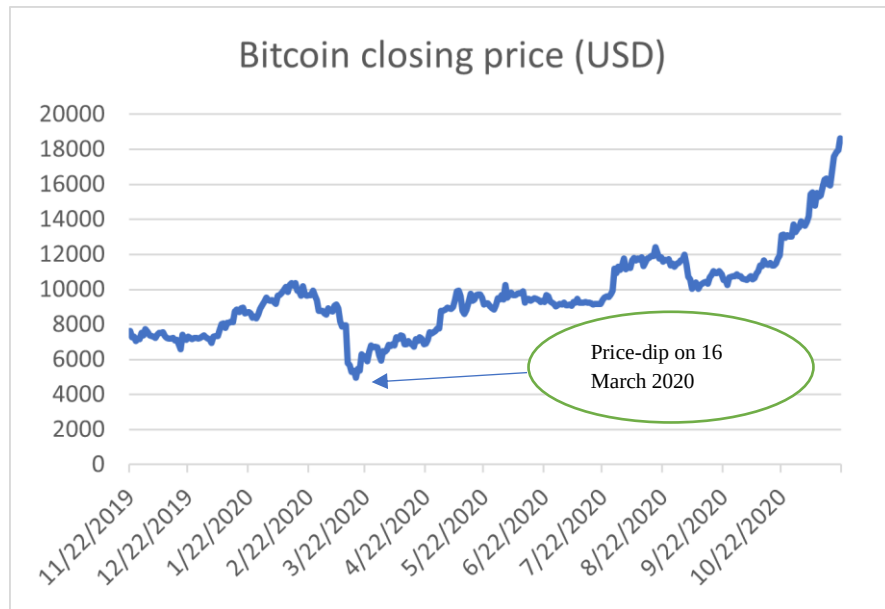
¹¹⁵ Daniel Oberhaus, ‘Coinbase is Being Sued for Insider Trading’ *The Vice* (05 March 2018) <https://www.vice.com/en_us/article/pam4xn/coinbase-insider-trading-lawsuit-gdax-bitcoin-cash> accessed 13 November 2020.

¹¹⁶ Stefan Stankovic, ‘US Cryptocurrency Regulation: Policies, Regimes & More’ *The Unblock* (18 February 2019) <<https://unblock.net/us-cryptocurrency-regulation/#h3>> accessed 21 November 2020.

¹¹⁷ CNN Business, <<https://edition.cnn.com/business/live-news/stock-market-news-today-031620/index.html>> accessed 21 November 2020.

¹¹⁸ Arjun Kharpal, ‘Over \$26 billion wiped off cryptocurrency market in 24 hours after massive oil price plunge’ *The CBBC* (08 March 2020) <https://www.cnn.com/2020/03/09/bitcoin-btc-and-other-cryptocurrency-prices-plunge-after-oil-drop.html?_source=sharebar|linkedin&par=sharebar> accessed 21 November 2020.

¹¹⁹ Jose Antonio Lanz, ‘Ethereum Price Dip Wipes \$1.5 Billion from DeFi Markets’ *The Decrypt* (31 October 2020) <<https://decrypt.co/46796/ethereum-price-1-5-billion-defi-markets>> accessed 21 November 2020.





Source: Coindesk

Figure 1: Daily closing prices of bitcoin, Ethereum, and ripple between November 22, 2019, and November 21, 2020.

In light of cryptocurrency's price volatility and regulatory opacity, the new addition of federally regulated cryptoderivatives to the market will give these investors complacency that their investments are protected under federal laws. However, as of now, the regulators' approach is somewhat "do your own research" and "take the risk on your own."¹²⁰ Therefore, in the context of an unregulated spot market and the CFTC's lack of oversight on it, investor protection in the cryptoderivatives market is very much questionable.

D. The SEC Contradicts the CFTC's View as the SEC Rejects ETFs on the Ground of Fraud and Abusive Practices in the Cryptocurrency' Spot' Market

Unlike the CFTC, the SEC appears to be reluctant to approve any new cryptocurrency products that would require oversight in the spot market.¹²¹ On several occasions, SEC Chairman Jay

¹²⁰ The SEC and the CFTC have been issuing customer warnings from time to time.

¹²¹ See Marion A. Brown, 'Cryptocurrency and Financial Regulation: The SEC's Rejection of Bitcoin-Based ETPs' [2012] 23 N.C. Banking Institute 139 (Discussing two-prong requirements under section 6(b)(5) of the Surveillance Agreement Test – the first prong is, defining 'regulated market;' and the second prong is defining 'significant size.' CSRO could break bitcoin ETPs out of the regulatory catch-22.). Also, Tom Lydon, 'SEC Rejects 9 Applications for Bitcoin ETFs' *The NASDAQ* (New York, 23 August 2018)

Clayton has raised concerns about the current cryptocurrency markets featuring less investor protection and more susceptibility to fraud and manipulation.¹²² The SEC's view became evident when it first rejected a bitcoin exchange-traded products ("ETPs")¹²³ in 2018.¹²⁴ To date, the SEC has disapproved of more than nine bitcoin ETP proposals, including bitcoin ETP proposals from Proshares,¹²⁵ Direxion,¹²⁶ and GraniteShares,¹²⁷ and the latest being the ETP application by Wilshire Phoenix.¹²⁸ In rejecting the bitcoin ETPs applications, the SEC held that none of the applicants had proved that the cryptocurrency market is uniquely resistant to market manipulation to secure investor protection and public interest as required under §6(b)(f) of the Exchange Act 1934. These ETP products are not designed to "prevent fraudulent and

<<https://www.nasdaq.com/articles/sec-rejects-9-applications-bitcoin-etfs-2018-08-23>> accessed 13 November 2020.

¹²² See SEC Chairman Jay Clayton, 'Statement on Cryptocurrencies and Initial Coin Offerings' (11 December 2017) <<https://www.sec.gov/news/public-statement/statement-clayton-2017-12-11>> accessed 14 November 2020 (Warning that cryptocurrency markets span national borders as the investment funds may travel across boundaries rapidly. As a result, "risks can be amplified, including the risk that market regulators, such as the SEC, may not be able to effectively pursue bad actors or recover funds." See also SEC Chairman Jay Clayton and CFTC Chairman Christopher Giancarlo, 'Statement by SEC Chairman Jay Clayton and CFTC Chairman J. Christopher Giancarlo: Regulators are Looking at Cryptocurrency' SEC (25 January 2018) <<https://www.sec.gov/news/public-statement/statement-clayton-giancarlo-012518>> accessed 13 November 2020.

¹²³ ETPs are securities that are traded on exchanges similar to stocks. Cryptocurrency ETPs could be in two basic forms: (1) ETFs holding cryptoderivatives, and (2) ETPs physically holding cryptocurrency. See James Chan, 'Exchange Traded Products (ETPs)' *The Investopedia* (25 August 2019) <<https://www.investopedia.com/terms/e/exchange-traded-products-etp.asp>> accessed 13 November 2020.

¹²⁴ Kate Rooney and Bob Pisani, 'Winklevoss twins bitcoin ETF rejected by SEC' *The CNBC* (26 July 2018) <<https://www.cnbc.com/2018/07/26/winklevoss-twins-bitcoin-etf-rejected-by-sec.html>> accessed 13 November 2020.

¹²⁵ SEC, 'SEC Release No. 82350 (19 December 2017) and SEC Release No. 82 FR 61100 (Dec. 26, 2017)' (proposing two ProShares bitcoin-futures ETPs).

¹²⁶ SEC, 'SEC Release Nos. 82532 (18 January 2018) and 83 FR 3380 (24 January 2018) (SR-NYSEArca-2018-02) (proposing five Direxion bitcoin-futures ETPs).

¹²⁷ SEC, SEC Release No. 34-83913 (22 August 2018) < <https://www.sec.gov/rules/sro/cboebzx/2018/34-83913.pdf>> accessed 13 November 2020. For commentary, see Nikhilesh De, 'The Securities and Exchange Commission (SEC) has issued rejections to bitcoin exchange-traded fund (ETFs) proposals from ProShares, Direxion and GraniteShares' *The Coindesk* (27 November 2019) <<https://www.coindesk.com/sec-rejects-7-bitcoin-etf-proposals>> accessed 13 November 2020.

¹²⁸ Nikhilesh De, 'SEC Rejects Latest Bitcoin ETF Bid' *The Coindesk* (27 February 2020) <<https://www.coindesk.com/sec-rejects-latest-bitcoin-etf-bid>> accessed 14 November 2020.

manipulative acts and practices,”¹²⁹ as the current bitcoin futures market (such as CMR and CBOE) is not of a significant size, which technically prevents the DCMs from detecting and deterring misconduct and trace price manipulation despite the use of an information-sharing agreement.¹³⁰ With regard to investor protection, although the SEC opined that “trading a bitcoin-based ETP on a national securities exchange might provide some additional protection to investors,”¹³¹ this protection is not sufficient to fulfill the requirements of §6(b)(f) of the Exchange Act that requires the “rules of a national securities exchange be designed to prevent fraudulent and manipulative acts and practices.”¹³²

Regarding these new cryptocurrency products, the SEC also raises the issue of custody risk.¹³³ As bitcoin is largely traded on unregulated international exchanges, these custodians carry a significant risk of being hacked or going out of business.¹³⁴ Custody risk is also present in the cryptoderivatives markets, which is rarely addressed by the CFTC. Particularly, for physically-settled bitcoin futures contracts, the exchanges are required to hold physical bitcoins.¹³⁵ Given that there is no federal-level investor protection for these trusts, bitcoin held by the trust is not

¹²⁹ See Katie Rooney and Bob Pisani, ‘Winklevoss Twins Bitcoin ETF Rejected by SEC’ *The CNBC* (26 July 2018) <<https://www.cnbc.com/2018/07/26/winklevoss-twins-bitcoin-etf-rejected-by-sec.html>> accessed 30 November 2019).

¹³⁰ Note 116 at 24. The rationale behind is that to successfully manipulate the ETP, one would also have to trade on the spot market. In traditional commodity a surveillance-sharing agreement assists the ETP listing markets in spotting manipulative behavior in the spot market.

¹³¹ Note 129 at 29.

¹³² *Id.*

¹³³ *Id.*

¹³⁴ Daniel Shane, ‘Bitcoin Exchange Goes Bust After Hack’ *The CNN Tech* (20 December 2017) <<https://money.cnn.com/2017/12/20/technology/south-korea-bitcoin-exchange-closes/index.html>> accessed 23 November 2020.

¹³⁵ A company’s CEO gone missing with Cold Wallet’s access. In another occasion the CEO died who only knew the password of the company’s cold wallet. In both cases, the investors lost all the cryptocurrencies and money as well. See Matthew Beedham, ‘Cryptocurrency Exchange IDAX’s CEO Reportedly Missing With Company’s Wallet’ *The Hard Fork* (29 November 2019) <<https://thenextweb.com/hardfork/2019/11/29/cryptocurrency-exchange-ceo-missing-idax-bitcoin-cold-wallet/>> accessed 14 November 2020; see also Antonia Noor Farzan, ‘Millions vanished with a cryptocurrency entrepreneur’s sudden death. Now investors want his body exhumed’ *The Washington Post* (16 December 2019) <<https://www.washingtonpost.com/nation/2019/12/16/gerald-cotten-quadrigacx-cryptocurrency-death-body-exhumed/>> accessed 14 November 2020.

subject to Federal Deposit Insurance Corporation or SIPA. Therefore, if cryptocurrencies are lost or stolen, or a crypto holder dies, it is likely that those coins will be lost forever.¹³⁶

E. The CFTC's Approach Deviates from the Major Global Regulators

The CFTC's approach in approving cryptoderivatives deviates from the other two major global regulators—the ESMA and the FCA. Both the EU and the UK regulators took measures restricting the trading of cryptoderivatives, determining that retail investors are not protected from the price volatility, speculation, and other forms of market and operational risks associated with cryptocurrencies.¹³⁷ Also, the “complexity of these products and a lack of transparency” limit retail investors' ability to understand the risks underlying these products.¹³⁸ Leveraged cryptoderivatives are risky and extremely volatile, increasing the scale and speed of investors' losses from a cryptoderivative.¹³⁹ Furthermore, the regulators should be cautious about the risk of the cryptocurrency speculative bubble.¹⁴⁰ Although many cryptoenthusiasts believe that cryptoderivatives could increase liquidity in the cryptocurrency market, thereby stabilizing price volatility,¹⁴¹ in the absence of a comprehensive regulation addressing the core regulatory concerns, cryptoderivatives would still hurt retail investors.

IV. **Future of Cryptoderivatives in the US: Possible Regulatory Frameworks**

In order to achieve a robust and effective cryptoderivatives regulatory framework in the US, it is essential that (1) the US cryptoderivatives market is free from manipulative and abusive

¹³⁶ Josias Dewey (ed.), 'Global Legal Insights- Blockchain & Cryptocurrency Regulation' [2019] *Global Legal Insights* 61 <https://www.acc.com/sites/default/files/resources/vl/membersonly/Article/1489775_1.pdf> accessed 16 November 2020.

¹³⁷ Id at pp.5-10.

¹³⁸ Id at p.51.

¹³⁹ Id.

¹⁴⁰ The bitcoin price is soaring as it hit \$18.8K on 20 November 2020. Joesph Young, 'Bitcoin Price Breaks Past \$18.8K as Bears and Bulls Map Out its Path' *The Cointelegraph* <<https://cointelegraph.com/news/bitcoin-price-breaks-past-18-8k-as-bears-and-bulls-map-out-its-path>> accessed 20 November 2020.

¹⁴¹ Many also view that having another competitor in the market or other altcoin derivatives could give a major boost to their liquidity and trading volumes. This may also create awareness of broader cryptocurrency market among investors which may result in infusing more money in the market. This could help create less volatility of altcoins' prices.

practices, (2) regulators have adequate visibility into the cryptocurrency spot market, (3) regulators are well-equipped to detect abusive and manipulative practices in the cryptoderivatives market, and (4) enforcement mechanisms are in place to safeguard investors interest. This paper explores two possible regulatory frameworks for cryptoderivatives. *First*, like the UK (and possibly the EU in the future), there could be a complete ban on cryptoderivatives enacted by a federal statute as cryptoderivatives are just a means of speculation, and the lack of oversight in the spot market will continue hurting retail investors. *Second*, if an outright ban is not feasible, Congress must develop comprehensive legislation (“cryptoregulation”) that recognizes the novel market and operational risks posed by cryptocurrency and introduces effective regulatory intervention in the cryptoderivatives markets.

A. The Possibility of a Complete Ban on Cryptoderivatives

Following the UK and the EU, the US regulators may consider banning the sale and purchase of cryptoderivatives in the derivatives exchanges keeping investor protection as their central focus. Cryptoderivatives pose a unique threat to investors due to their high leverage and extreme price volatility.¹⁴² Moreover, the failure of the CFTC to have any oversight mechanism on the cryptocurrency spot market is contrary to the CFTC’s mandate to protect investors from fraud and abusive market practices. Unless the CFTC has established a mechanism to oversee the cryptocurrency spot market meaningfully and adequately, a ban on cryptoderivatives will act as a warning to investors not to put their money in such risky products. Further, the complexity of the cryptoderivatives and the lack of transparency in the cryptocurrency spot market require a more rigorous enforcement approach from the CFTC. However, many argue that an outright ban is likely to hurt the existing cryptocurrency platforms that comply with the laws and regulations.¹⁴³ There is another set of arguments:

¹⁴² See Ryan Clements, ‘Cryptocurrency Self-Regulatory Organization (CSRO)’ (The FinReg Blog, 21 June 2019) <<https://sites.duke.edu/thefinregblog/2018/06/21/can-a-cryptocurrency-self-regulatory-organization-work-assessing-its-promise-and-likely-challenges/>> accessed 23 November 2020.

¹⁴³ Osato Avan-Nomayo ‘Cryptoderivatives Ban is Not a Good Idea’ *The Blockonomi* (08 October 2019), <<https://blockonomi.com/crypto-derivatives-ban-not-good-idea-says-wfe/>> accessed 23 November 2020.

“where regulator erred on the side of banning or bashing cryptocurrencies, they have faced classical problems of regulatory competition and regulatory arbitrage, i.e., the migration of the industry from their jurisdiction to more welcoming ones or migration of activities to underground or black-markets.”¹⁴⁴

Given the drawbacks of an outright ban on cryptoderivatives, this paper proposes an alternative regulatory framework—enacting comprehensive federal-level cryptoregulation in response to the emerging issues of manipulation and lack of investor protection in the cryptocurrency spot market and vis-à-vis cryptoderivatives markets.

B. The Need for a Comprehensive Federal Cryptoregulation

The need for a comprehensive cryptoregulation is premised on four grounds: *first*, the sporadic regulatory efforts among different US regulatory agencies concerning cryptocurrency are counterproductive. A systemic regulatory approach can minimize the risks of cryptocurrency spot markets and avert market failure.¹⁴⁵ *Second*, the novelty involved in cryptocurrency requires a uniform regulatory approach; otherwise, it may bring about an unwanted disruption in the capital and financial market. In the US, the regulatory approach to cryptocurrency is fragmented. For instance, while the SEC has declared ICO as a security, it did not establish its jurisdiction exclusively on all digital tokens. Meanwhile, the US Internal Revenue Service (IRS) considers convertible cryptocurrency as property for tax purposes.¹⁴⁶ In addition, the federal courts in several cases (such as *United States v. Ulbricht* ¹⁴⁷) have treated cryptocurrency as money for specific purposes. However, there are still divided opinions as to the status of cryptocurrency as “money” because (1) cryptocurrency is not widely accepted as a means of payment, and (2) its store value is unreliable due to market volatility.¹⁴⁸ It is argued

¹⁴⁴ Hossein Nabilou, ‘How to Regulate Bitcoin? Decentralized Regulation for a Decentralized Cryptocurrency’ [2019] 27 International Journal on Law and Information Technology 266.

¹⁴⁵ Id.

¹⁴⁶ IRS, ‘Virtual Currencies’ <<https://www.irs.gov/businesses/small-businesses-self-employed/virtual-currencies>> accessed 21 November 2020.

¹⁴⁷ No. 15-1815 (2d Cir. 2017). Retrieved from <https://cases.justia.com/federal/appellate-courts/ca2/15-1815/15-1815-2017-05-31.pdf?ts=1496241010>.

¹⁴⁸ S&P Global <<https://www.spglobal.com/en/research-insights/articles/the-future-of-banking-cryptocurrencies-will-need-some-rules-to-change-the-game>> (accessed Dec. 4, 2019) states: “cryptocurrencies do not meet the basic two requisites of a currency: An effective mean of exchange and an effective store of value. First, cryptocurrencies are still not

that a comprehensive federal cryptoregulation can cure the problem of this fragmented, sporadic, and ambiguous regulatory approach by adopting a unanimous definition of cryptocurrency (or by defining it as a separate ‘digital asset class’) and thus bringing the intermediaries and cryptocurrency-based assets under the same regulatory framework.¹⁴⁹ *Third*, if fraud and market manipulation continue, it will eventually drive potential investors away from the market. Further, opacity and lack of regulatory clarity can result in the loss of investors’ confidence.¹⁵⁰ *Finally*, if cryptocurrencies become an effective monetary instrument in the future, their impact on the country’s monetary policy would be profound as the Federal Reserve Board might lose its ability to control the money supply.¹⁵¹ Therefore, there is a demand from both policymakers and market participants that Congress develop comprehensive federal legislation to regulate cryptocurrencies¹⁵² and bring them under a uniform legislative scope.¹⁵³

In an ongoing effort to achieve regulatory clarities, in March 2020, the US Senate proposed a new Cryptocurrency Act 2020.¹⁵⁴ The draft bill categorized cryptocurrency or digital tokens into three main groups based on its decentralized nature and/or use of cryptographic ledger—

widely accepted as payment instruments, although the list of companies accepting them have increased over the past few years. Second, the volatility that we have observed over the past 12 months in the valuation of some cryptocurrencies and their market cap is the most meaningful evidence that they fail the test of value storage. We also don't view cryptocurrencies as an asset class. For starters, the total outstanding aren't big enough yet. At Feb. 10, 2018, there were 1,523 outstanding cryptocurrencies with a market cap of around \$394 billion. By way of comparison, at the same date, this is well below the market capitalization of Apple Inc., around \$794 billion.” (Charts omitted).

¹⁴⁹ Averie Brooks, ‘U.S. Regulation of Blockchain Currencies: A Policy Overview’ [2018] 9 (2) Amherst Intellectual Property Brief 75 <http://www.ipbrief.net/wp-content/uploads/2018/05/USRegulationBlockchainCurrencies_FINAL.pdf> accessed 20 November 2020.

¹⁵⁰ *Id.*

¹⁵¹ Mohammad Damak, The Future of Banking: Cryptocurrencies Will Need Some Rules to Change the Game, S&P GLOBAL (19 February 2018), <<https://www.spglobal.com/en/research-insights/articles/the-future-of-banking-cryptocurrencies-will-need-some-rules-to-change-the-game>> accessed 19 November 2020.

¹⁵² In the 2019 Annual Report, the Financial Stability Oversight Council (FSOC) called for strict state and federal regulations of stablecoins and digital assets. FSOC, ‘Annual Report’ (2019) <<https://home.treasury.gov/system/files/261/FSOC2019AnnualReport.pdf>> accessed 20 November 2020.

¹⁵³ Peter Van Valkenburgh, ‘A national alternative to onerous state-by-state regulation of cryptocurrency intermediaries’, *The Coin Center* (30 August 2019) <<https://coincenter.org/entry/a-national-alternative-to-onerous-state-by-state-regulation-of-cryptocurrency-intermediaries>> accessed 20 November 2020.

¹⁵⁴ [Text - H.R.6154 - 116th Congress \(2019-2020\): Crypto-Currency Act of 2020 | Congress.gov | Library of Congress.](#)

(1) cryptocurrencies including the US currency representation; (2) crypto-commodities residing on a blockchain or decentralized cryptographic ledger; and (2) crypto-securities that meet *Howey* test.¹⁵⁵ The Act further proposed that depending on the categories, the CFTC, the SEC, and the FinCEN would have regulatory authorities over this asset class. In another draft bill, ‘Digital Commodity Exchange Act 2020’,¹⁵⁶ The US House of Representatives proposed an amendment to the CEA by incorporating definitions of ‘digital commodity’,¹⁵⁷ ‘digital commodity custodian’,¹⁵⁸ and ‘digital commodity exchange’,¹⁵⁹ It recommended CFTC’s exclusive jurisdiction over digital commodity transaction. The proposition is based on the assumption that cryptocurrencies are a distinct form of digital-asset class classified as ‘virtual commodities.’

Nonetheless, both propositions do not adequately answer the regulatory quandaries regarding ICOs, cryptocurrency as a mode of payment, or property for tax purposes in as much as they do not complement the existing fragmented regulatory approach towards cryptocurrency. To provide a degree of regulatory clarities, this paper proposes a federal-level cryptoregulation with a separate regulatory agency that will have exclusive jurisdiction over cryptocurrencies in the US, including the spot market and any financial instruments where the underlying asset is a cryptocurrency (such as cryptoderivatives), and mandatory registration requirement for cryptocurrency exchanges.¹⁶⁰ The agency will also coordinate with other regulatory agencies as the new agency will work parallelly with the others (such as SEC, CFTC, IRS, and FinCEN) but will only be limited to regulating cryptocurrency and other digital assets. A uniform

¹⁵⁵ For reference to *Howey* test, see note 21.

¹⁵⁶ H.R. 8373: Digital Commodity Exchange Act of 2020. Retrieved from <<https://www.govinfo.gov/content/pkg/BILLS-116hr8373ih/pdf/BILLS-116hr8373ih.pdf>> accessed 23 November 2020.

¹⁵⁷ The proposed amendment defines ‘digital commodity’ as “any form of fungible intangible personal property that can be exclusively possessed and transferred person to person without necessary reliance on an intermediary, and which does not represent a financial interest in a company, partnership, or investment vehicle.” *Id.*

¹⁵⁸ “The term ‘digital commodity custodian’ means an entity that holds, maintains, or safeguards digital commodities and other assets on behalf of digital commodity market participants.” Note 168 at p.2.

¹⁵⁹ “The term ‘digital commodity exchange’ means a trading facility that lists for one digital commodity.” *Id.*

¹⁶⁰ The proposed new legislation should also exempt the SEC and the CFTC from exercising its jurisdiction over cryptocurrency-based financial instruments.

federal-level cryptocurrency agency is likely to establish effective oversight and supervisory authority over the cryptocurrency spot markets, which would help curb market manipulation and restore investor confidence.¹⁶¹

C. The Proposed Framework of the Cryptoregulation

The proposed cryptoregulation should be based on information, equal access, and investor confidence.¹⁶² In particular, it should have the mandate of protecting investors against fraud and offer a degree of centralization. Therefore, the regulation should have (1) a new federal cryptocurrency agency established by an Act of Congress; (2) the mandatory registration requirement for all cryptocurrency exchanges, including the cryptocurrency spot markets; and (3) a national cryptocurrency exchange.

i. *A new federal cryptocurrency agency: A de novo cryptoregulatory regime established by an Act of Congress*

A federal cryptocurrency agency having exclusive jurisdiction over cryptocurrencies can prevent price manipulation, fraud, and abusive market practices by establishing a direct oversight over the cryptocurrency intermediaries, including exchanges and spot markets, and thereby enhance investors' protection. Though the new cryptocurrency agency structure is subject to rigorous academic, technical, and regulatory discussions, it is not uncommon in the US to constitute a new federal agency to protect consumers and fill in the regulatory vacuum. After the financial crisis of 2008, the Dodd-Frank Wall Street Reform and Consumer Protection Act established the Consumer Financial Protection Bureau—a single, independent consumer-focused regulatory regime consolidating the scattered financial authorities throughout the federal government and bringing them under one roof.¹⁶³ Similarly, the *de novo* cryptoregulator should combine existing regulators' mandates, jurisdictions, responsibilities,

¹⁶¹ This paper does not propose a direct regulation of the cryptocurrency technology (i.e., blockchain technology) itself. Rather, the regulation should be indirect, meaning it should address the regulation of the cryptocurrency intermediaries, including cryptocurrency exchanges and custodians.

¹⁶² *Id.*

¹⁶³ Title X—Bureau of Consumer Financial Protection, the Dodd Frank Act. Retrieved from https://www.law.cornell.edu/wex/dodd-frank_title_x_-_bureau_of_consumer_financial_protection# accessed 20 November 2020.

and enforcement authorities over cryptocurrencies and overcome the current regulatory overlap and ambiguity. The *de novo* regime’s mandate should be “investor protection through promoting market transparency and preventing and policing fraud, market manipulation, and insider trading.

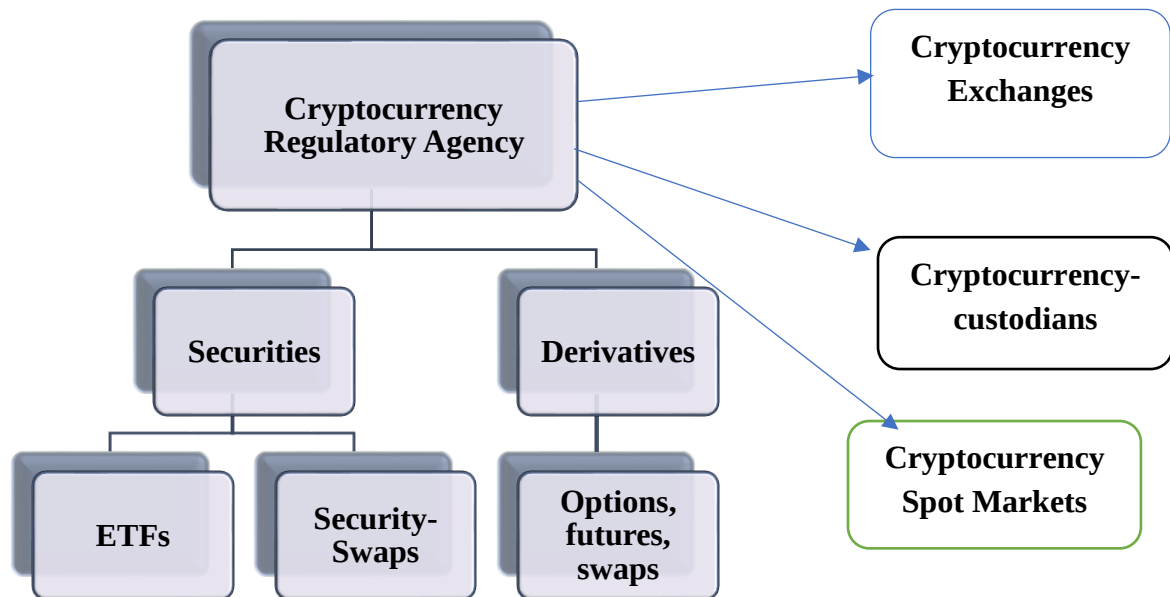


Figure 2: The Cryptocurrency Regulatory Agency’s Jurisdiction over Cryptocurrency-Based Financial Instruments

With respect to jurisdictions, this paper proposes that the cryptoregulator should deal with cryptocurrency-based securities (e.g., ICOs, digital tokens, utilities, etc.), derivatives (swaps, futures, and options), ETFs, cryptosecurities-based swaps, cryptocurrency custodians, and cryptocurrency spot markets. **(Figure 2)**

ii. *Federal licensing requirement: Overseeing the Cryptocurrency Spot Market*

The new regulator should be responsible for authorizing licenses to eligible and trusted cryptocurrency intermediaries, i.e., cryptocurrency exchanges, cryptocurrency custodians, payment processors, and cryptocurrency spot markets that will operate businesses in

cryptocurrencies. The licensing regime is significant for cryptocurrency companies because it will help the regulator bring the licensed entities under a single supervisory authority. For instance, after the infamous Mt. Gox¹⁶⁴ hacking incident, Japan enacted an amendment to the Payment Services Act (PSA) in 2017 and created a licensing regime for its cryptocurrency exchanges.¹⁶⁵ According to the new amendment, all cryptocurrency exchanges engaged in purchasing, selling, and exchanging cryptocurrencies and intermediaries thereof must register with Japan Financial Services Authority (FSA).¹⁶⁶ The exchanges must also comply with capital and positive net assets requirement and have an internal auditing system to ensure compliance with the relevant PSA rules.¹⁶⁷ To date, the FSA has granted licenses to 16 cryptocurrency exchanges subjecting them to regulatory oversight.¹⁶⁸ Like Japan, the licensing regime will help the US regulator supervise the cryptocurrency intermediaries' and establish a market oversight mechanism by ensuring that these intermediaries comply with the relevant US laws, including AML/CFT, anti-manipulation, anti-fraud, consumer disclosure, and prudential (licensing, minimum-capitalization) requirements.

iii. *A Central Cryptocurrency Trading Platform with Registration Requirement*

Under the new cryptocurrency regulation, there should be a central cryptocurrency trading platform with mandatory registration requirements for all cryptocurrency exchanges and spot markets willing to trade cryptocurrencies and issue cryptocurrency-based offerings. A centrally regulated cryptocurrency trading platform with a mandatory registration requirement will

¹⁶⁴ In 2014, Bitcoin hacking bankrupted a leading exchange in Japan called Mt. Gox, with approximately half-a-billion dollars in bitcoin (\$850,000 bitcoin) stolen. While \$200,000 bitcoin were recovered within six months, their dollar value sunk by the revelation of weak security, and the incident showed that hacks impact bitcoin's trading price. Exchange customers had no remedy. Robert McMillan, *The Inside Story of Mt. Gox, Bitcoin's \$460 Million Disaster*, WIRED (Mar. 3, 2014), www.wired.com/2014/03/bitcoin-exchange/.

¹⁶⁵ Masahiko Ishida, Edward Mears, and Ryutaro Takeda, 'Japan Regulatory Update on Virtual Currency Business' *DLA Piper* (29 December 2017) <<https://www.dlapiper.com/en/japan/insights/publications/2017/12/japan-regulatory-update-on-virtual-currency-business/>> accessed 23 November 2020.

¹⁶⁶ *Id.*

¹⁶⁷ *Id.*

¹⁶⁸ These cryptocurrency exchanges are represented by the Japan Virtual Exchange Association—a self-regulatory organization for the Japanese cryptocurrency industry.

infuse the liquidity in the spot market and stabilize the cryptocurrency's price volatility.¹⁶⁹ This will further facilitate the regulators to exercise direct oversight over the cryptocurrency spot markets. A central platform predicated on disclosure requirements and information-sharing will provide investors with transparency and equal access to information and will safeguard the market from the cryptocurrency world, which is highly asymmetrical, unverified, and sometimes blatantly false.¹⁷⁰ As to the platform's structure, the new agency could create and maintain a central database where all cryptocurrency-based offerings will be listed and conducted, providing the regulator "with direct oversight, capabilities, and transparency for all transactions and parties involved."¹⁷¹ Further, to prevent false claims on ICOs, cryptoderivatives, and other cryptocurrency-based offerings, there should be a requirement that all advertisements and solicitations are source-verified.¹⁷² The registration and offering process should be based on disclosure requirements, representations, and warranties, as well as the regulator's involvement at the outset. In addition to supporting, facilitating, and listing ICOs and cryptoderivatives, a central cryptocurrency exchange could employ world-class security measures, which would consequently enhance the trading platform's safety and investors' confidence.¹⁷³

V. Conclusion

There are many questions yet to be answered in this rapidly changing cryptocurrency industry. However, hype, volatility, speculation—these three words best describe the current cryptocurrency market. The sporadic and fragmented regulatory responses to cryptoderivatives demonstrate the dire need for a uniform and comprehensive cryptocurrency regulation in the US. The lack of regulation perpetuates the fraudulent and manipulative behavior in the spot market, which will eventually drive potential investors away from the market. Many crypto

¹⁶⁹ Brial Novell, 'Regulation Crypto—A Proposed Framework for United States Cryptocurrency Regulation' 20 August 2018 <<https://www.prnewswire.com/news-releases/regulation-crypto--a-proposed-framework-for-united-states-cryptocurrency-regulation-300699235.html>> accessed 23 November 2020.

¹⁷⁰ Id.

¹⁷¹ Id.

¹⁷² Id.

¹⁷³ Id.

enthusiasts resist the idea of regulating the industry on the ground that it may impede innovation. Yet regulatory clarities and a proper oversight mechanism over the cryptocurrency exchanges and spot markets are necessary frictions as this will help protect the market integrity, punish abusive and manipulative practices, and restore investors' confidence. Therefore, Congress should develop comprehensive federal cryptocurrency legislation that will capture cryptocurrency's novelty and its underpinning technology. A comprehensive regulation will serve the public interest providing a systemic regulatory approach that minimizes the risks of cryptocurrency spot markets and averts market failure. Further, the legislation will encapsulate the technology-specific regulation to cryptocurrency strengthening the US capital and financial market. A uniform federal-level cryptocurrency agency is likely to establish effective oversight and supervisory authority over the cryptocurrency spot markets, which would help curb market manipulation and protect retail investors.

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